

Falcon Post-Quantum Signature Scheme

Function-Level Profiling Report
-00 vs -03 · 1000 iterations · gprof

Falcon-512 / Falcon-1024

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Contents

1	Overview	2
2	Runtime Summary: O0 vs O3	2
3	Category Weight per Operation	2
3.1	Category Breakdown — -00	2
3.2	Category Breakdown — -03	3
4	Per-Operation Function Detail	3
4.1	Falcon-512 — KeyGen	3
4.2	Falcon-512 — Sign	6
4.3	Falcon-512 — Verify	9
4.4	Falcon-1024 — KeyGen	11
4.5	Falcon-1024 — Sign	15
4.6	Falcon-1024 — Verify	18
5	Key Findings	20

1 Overview

This report presents a function-level profiling analysis of the **Falcon** post-quantum digital signature scheme (NIST PQC Round 3 finalist, lattice-based). Two parameter sets are compared: **Falcon-512** (NIST Level 1) and **Falcon-1024** (NIST Level 5). Each operation was executed **1000** times and profiled with **gprof** under two compiler settings: `-O0 -pg` and `-O3 -pg`.

Functions are grouped into three categories:

- **Hash / PRNG** — SHAKE-256 (Keccak), PRNG refill, hash-to-point.
- **Integer Arithmetic** — Modular NTT (`modp_*`, `mq_*`), multi-precision integers (`zint_*`), Gaussian sampler (`mkgauss`).
- **Floating-Point / FFT** — Custom 53-bit FPR type (`fpr_*`, `FPR`), FFT/iFFT, polynomial operations.

2 Runtime Summary: O0 vs O3

Table 1: Total cumulative runtime (seconds, 1000 iterations)

Variant	Operation	O0 (s)	O3 (s)	Speedup
Falcon-512	KeyGen	52.35	12.37	4.23 ×
Falcon-512	Sign	16.81	3.24	5.19 ×
Falcon-512	Verify	0.17	0.03	5.67 ×
Falcon-1024	KeyGen	154.41	34.26	4.51 ×
Falcon-1024	Sign	36.09	7.57	4.77 ×
Falcon-1024	Verify	0.35	0.15	2.33 ×

3 Category Weight per Operation

The following tables show what fraction of total profiled time is spent in each functional category for O0 and O3 builds.

3.1 Category Breakdown — -O0

Table 2: Category time share -O0 (%)

Variant	Operation	Hash	Integer	FP/FFT	Other
Falcon-512	KeyGen	5.9%	32.3%	60.5%	1.3%
Falcon-512	Sign	1.3%	2.9%	95.4%	0.4%
Falcon-512	Verify	17.6%	82.4%	0.0%	0.0%
Falcon-1024	KeyGen	3.0%	33.5%	62.5%	1.0%
Falcon-1024	Sign	1.6%	1.8%	96.3%	0.4%
Falcon-1024	Verify	17.1%	80.0%	2.9%	0.0%

Table 3: Category time share -O3 (%)

Variant	Operation	Hash	Integer	FP/FFT	Other
Falcon-512	KeyGen	7.8%	37.9%	52.2%	2.0%
Falcon-512	Sign	1.5%	2.5%	94.8%	1.2%
Falcon-512	Verify	33.3%	66.7%	0.0%	0.0%
Falcon-1024	KeyGen	4.2%	40.7%	54.1%	1.1%
Falcon-1024	Sign	1.1%	3.2%	95.5%	0.3%
Falcon-1024	Verify	20.0%	73.3%	0.0%	6.7%

3.2 Category Breakdown — -O3

4 Per-Operation Function Detail

Each subsection shows the top functions per category for both O0 and O3 builds. Column headers: % = percentage of total profiled time, **self(s)** = self seconds, **calls** = call count.

4.1 Falcon-512 — KeyGen

-O0 build (total: 52.350 s)

Hash / PRNG — 5.9% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	3.93	2.0600	2.7M
keccak_inc_squeeze	1.57	0.8200	45.6M
shake256_inc_squeeze	0.23	0.1200	45.6M
get_rng_u64	0.20	0.1000	45.6M
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 32.3% of total

Function	%	self(s)	calls
modp_montymul	12.99	6.8000	982.1M
modp_add	4.63	2.4200	688.0M
zint_mod_small_unsigned	2.30	1.2100	50.1M
mkgauss	2.25	1.1800	11.4M
zint_add_scaled_mul_small	2.02	1.0600	18.2M
modp_NTT2_ext	1.32	0.6900	2.0M
modp_R2	1.06	0.5600	3.8M
modp_sub	0.88	0.4600	368.7M

Function	%	self(s)	calls
zint_add_mul_small	0.83	0.4300	21.7M
zint_finish_mod	0.78	0.4100	893.2K
zint_norm_zero	0.53	0.2800	21.2M
modp_iNTT2_ext	0.40	0.2100	1.8M
zint_mod_small_signed	0.35	0.1800	28.3M
mq_montymul	0.29	0.1500	18.3M
zint_bezout	0.29	0.1500	1.0K

Floating-Point / FFT — 60.5% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_fpr_add	30.13	15.7800	301.5M
FPR	11.44	5.9900	662.4M
PQCLEAN_FALCON512_CLEAN_fpr_mul	10.12	5.3000	305.7M
PQCLEAN_FALCON512_CLEAN_fpr_scaled	3.95	2.0700	53.3M
fpr_ulsh	1.77	0.9300	301.5M
fpr_ursh	1.54	0.8100	301.5M
PQCLEAN_FALCON512_CLEAN_fpr_div	0.48	0.2500	2.0M
PQCLEAN_FALCON512_CLEAN_iFFT	0.32	0.1700	304.4K
PQCLEAN_FALCON512_CLEAN_FFT	0.31	0.1600	610.5K
fpr_sub	0.18	0.1000	120.8M
fpr_rint	0.08	0.0400	4.3M
PQCLEAN_FALCON512_CLEAN_poly_mul_fft	0.06	0.0300	580.2K
fpr_ulsh	0.04	0.0200	4.3M
fpr_neg	0.03	0.0100	8.1M
fpr_of	0.00	0.0000	53.3M

Other — 1.3% of total

Function	%	self(s)	calls
poly_big_to_fp	0.27	0.1400	596.3K
modp_div	0.19	0.1000	1.0M
poly_sub_scaled	0.17	0.0900	522.0K
modp_mkgm2	0.13	0.0700	1.0M
poly_sub_scaled_ntt	0.09	0.0400	56.1K
make_fg_step	0.08	0.0400	44.4K
solve_NTRU_intermediate	0.08	0.0400	7.0K
modp_ninv31	0.06	0.0300	2.8M
fpr_lt	0.05	0.0300	6.5M
modp_R	0.03	0.0100	6.8M
modp_Rx	0.02	0.0100	968.0K
mq_div_12289	0.02	0.0100	529.2K
poly_small_to_fp	0.02	0.0100	11.3K
make_fg	0.02	0.0100	8.1K
PQCLEAN_FALCON512_CLEAN_trim_i8_encode	0.02	0.0100	3.0K

-03 build (total: 12.370 s)

Hash / PRNG — 7.8% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	4.85	0.6000	2.7M
get_rng_u64	1.70	0.2100	45.4M
keccak_inc_finalize	1.29	0.1600	45.4M
shake256_inc_squeeze	0.00	0.0000	45.4M
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 37.9% of total

Function	%	self(s)	calls
modp_montymul	16.98	2.1000	967.8M
mkgauss	2.99	0.3700	11.4M
zint_rebuild_CRT.constprop.2	2.75	0.3400	36.2M
zint_mod_small_unsigned	2.43	0.3000	49.5M
zint_add_scaled_mul_small	2.43	0.3000	18.1M
modp_add	2.10	0.2600	685.7M
modp_sub	1.54	0.1900	366.7M
zint_add_mul_small	1.46	0.1800	21.7M
zint_finish_mod	1.13	0.1400	894.1K
modp_iNTT2_ext	0.97	0.1200	1.8M
zint_co_reduce_mod	0.57	0.0700	447.0K
poly_big_to_small	0.40	0.0500	20.1M
modp NTT2_ext	0.40	0.0500	220.2K
zint_bezout	0.32	0.0400	1.0K
zint_norm_zero	0.24	0.0300	20.1M

Floating-Point / FFT — 52.2% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_fpr_add	27.08	3.3500	302.6M
PQCLEAN_FALCON512_CLEAN_fpr_mul	12.93	1.6000	306.4M
FPR	3.23	0.4000	664.3M
PQCLEAN_FALCON512_CLEAN_fpr_scaled	2.91	0.3600	53.3M
fpr_trunc	1.70	0.2100	302.6M
fpr_ursh	1.54	0.1900	302.6M
PQCLEAN_FALCON512_CLEAN_fpr_div	0.97	0.1200	2.0M
PQCLEAN_FALCON512_CLEAN_iFFT	0.81	0.1000	303.7K

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_FFT	0.49	0.0600	609.1K
fpr_sub	0.32	0.0400	121.3M
fpr_ulsh	0.08	0.0100	4.3M
fpr_ursh	0.08	0.0100	4.3M
PQCLEAN_FALCON512_CLEAN_poly_mulconst	0.08	0.0100	11.3K
fpr_of	0.00	0.0000	53.3M
fpr_neg	0.00	0.0000	8.1M

Other — 2.0% of total

Function	%	self(s)	calls
poly_big_to_fp	0.81	0.1000	590.7K
modp_mkgm2	0.24	0.0300	995.0K
poly_sub_scaled	0.24	0.0300	520.6K
modp_div	0.16	0.0200	995.0K
solve_NTRU_intermediate	0.16	0.0200	7.0K
modp_ninv31	0.08	0.0100	2.8M
solve_NTRU_binary_depth1	0.08	0.0100	1.5M
poly_sub_scaled_ntt	0.08	0.0100	56.1K
make_fg_step.constprop.0	0.08	0.0100	36.2K
poly_big_to_fp.constprop.0	0.08	0.0100	2.0K
mq_montysqr	0.00	0.0000	5.8M
modp_R	0.00	0.0000	5.8M
fpr_inv	0.00	0.0000	2.0M
mq_conv_small	0.00	0.0000	1.1M
modp_Rx	0.00	0.0000	965.8K

4.2 Falcon-512 — Sign

-00 build (total: 16.810 s)

Hash / PRNG — 1.3% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_prng_refill	1.19	0.2000	39.2K
prng_get_u8	0.06	0.0100	5.4M
PQCLEAN_FALCON512_CLEAN_hash_to_point_ct	0.06	0.0100	1.0K
prng_get_u64	0.00	0.0000	1.8M
keccak_inc_squeeze	0.00	0.0000	718.0K
shake256_inc_squeeze	0.00	0.0000	718.0K
KeccakF1600_StatePermute	0.00	0.0000	12.0K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
keccak_inc_finalize	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K

Function	%	self(s)	calls
shake256_inc_ctx_release	0.00	0.0000	2.0K
shake256_inc_finalize	0.00	0.0000	2.0K

Integer Arithmetic — 2.9% of total

Function	%	self(s)	calls
mq_montymul	1.19	0.2000	21.0M
PQCLEAN_FALCON512_CLEAN_gaussian0_sample	0.80	0.1400	1.8M
mq NTT	0.36	0.0600	3.0K
mq_add	0.24	0.0400	9.2M
PQCLEAN_FALCON512_CLEAN_sampler	0.12	0.0200	1.0M
mq_iNTT	0.12	0.0200	1.0K
mq_sub	0.00	0.0000	9.2M

Floating-Point / FFT — 95.4% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_fpr_add	48.33	8.1100	148.4M
FPR	17.82	2.9900	297.1M
PQCLEAN_FALCON512_CLEAN_fpr_mul	13.86	2.3300	133.8M
fpr_ulsh	3.25	0.5500	148.4M
PQCLEAN_FALCON512_CLEAN_fpr_scaled	3.16	0.5300	12.0M
fpr_ursh	2.29	0.3900	152.0M
PQCLEAN_FALCON512_CLEAN_fpr_div	1.34	0.2300	2.3M
PQCLEAN_FALCON512_CLEAN_fpr_sqrt	1.19	0.2000	512.0K
PQCLEAN_FALCON512_CLEAN_fpr_expm_p63	0.95	0.1600	1.8M
PQCLEAN_FALCON512_CLEAN_poly_LDL_fft	0.54	0.0900	511.0K
fpr_trunc	0.48	0.0800	3.6M
PQCLEAN_FALCON512_CLEAN_FFT	0.42	0.0700	9.0K
fpr_sub	0.36	0.0600	70.4M
PQCLEAN_FALCON512_CLEAN_poly_split_fft	0.30	0.0500	2.0M
fpr_neg	0.18	0.0300	14.1M

Other — 0.4% of total

Function	%	self(s)	calls
mq_div_12289	0.18	0.0300	512.0K
PQCLEAN_FALCON512_CLEAN_comp_encode	0.12	0.0200	1.0K
fpr_floor	0.06	0.0100	1.0M
BerExp	0.03	0.0100	1.8M
mq_montysqr	0.00	0.0000	5.6M
fpr_inv	0.00	0.0000	2.3M
mq_conv_small	0.00	0.0000	1.5M
fpr_irsh	0.00	0.0000	1.0M
mq_rshift1	0.00	0.0000	9.0K
smallints_to_fpr	0.00	0.0000	8.0K

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_trim_i8_decode	0.00	0.0000	3.0K
PQCLEAN_FALCON512_CLEAN_complete_private	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_crypto_sign_signature	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_is_short_half	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_sign_dyn	0.00	0.0000	1.0K

-03 build (total: 3.240 s)

Hash / PRNG — 1.5% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_hash_to_point_ct	0.62	0.0200	1.0K
prng_get_u64	0.31	0.0100	1.8M
keccak_inc_finalize	0.31	0.0100	720.0K
PQCLEAN_FALCON512_CLEAN_prng_refill	0.31	0.0100	39.1K
prng_get_u8	0.00	0.0000	5.3M
shake256_inc_squeeze	0.00	0.0000	718.0K
KeccakF1600_StatePermute	0.00	0.0000	12.0K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K
shake256_inc_ctx_release	0.00	0.0000	2.0K
shake256_inc_finalize	0.00	0.0000	2.0K
shake256_inc_init	0.00	0.0000	2.0K

Integer Arithmetic — 2.5% of total

Function	%	self(s)	calls
mq_montymul	1.23	0.0400	20.0M
PQCLEAN_FALCON512_CLEAN_gaussian0_sampler	0.62	0.0200	1.8M
mq_add	0.31	0.0100	9.2M
PQCLEAN_FALCON512_CLEAN_sampler	0.31	0.0100	1.0M
mq_sub	0.00	0.0000	9.2M
mq_montymul.constprop.0	0.00	0.0000	1.0M
mq NTT	0.00	0.0000	3.0K
mq_iNTT	0.00	0.0000	1.0K

Floating-Point / FFT — 94.8% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_fpr_add	49.07	1.5900	148.4M
PQCLEAN_FALCON512_CLEAN_fpr_mul	19.44	0.6300	133.8M
FPR	7.10	0.2300	297.0M
PQCLEAN_FALCON512_CLEAN_fpr_div	4.63	0.1500	2.3M

Function	%	self(s)	calls
fpr_trunc	3.40	0.1100	152.0M
fpr_ursh	2.16	0.0700	152.0M
PQCLEAN_FALCON512_CLEAN_fpr_expm_p63	2.16	0.0700	1.8M
PQCLEAN_FALCON512_CLEAN_fpr_scaled	1.23	0.0400	12.0M
PQCLEAN_FALCON512_CLEAN_poly_LDL_fft	1.23	0.0400	511.0K
fpr_sub	0.62	0.0200	70.4M
fpr_rint	0.62	0.0200	1.0M
PQCLEAN_FALCON512_CLEAN_fpr_sqrt	0.62	0.0200	512.0K
PQCLEAN_FALCON512_CLEAN_FFT	0.62	0.0200	9.0K
fpr_half	0.31	0.0100	16.4M
fpr_sub	0.31	0.0100	6.4M

Other — 1.2% of total

Function	%	self(s)	calls
BerExp	0.31	0.0100	1.8M
mq_conv_small	0.31	0.0100	1.5M
mq_div_12289	0.31	0.0100	512.0K
PQCLEAN_FALCON512_CLEAN_complete_private	0.31	0.0100	1.0K
mq_montysqr	0.00	0.0000	5.6M
fpr_inv	0.00	0.0000	2.3M
fpr_floor	0.00	0.0000	1.0M
fpr_irsh	0.00	0.0000	1.0M
mq_rshift1	0.00	0.0000	9.0K
smallints_to_fpr	0.00	0.0000	8.0K
PQCLEAN_FALCON512_CLEAN_trim_i8_decode	0.00	0.0000	3.0K
PQCLEAN_FALCON512_CLEAN_comp_encode	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_crypto_sign_signature	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_is_short_half	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_sign_dyn	0.00	0.0000	1.0K

4.3 Falcon-512 — Verify

-00 build (total: 0.170 s)

Hash / PRNG — 17.6% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_hash_to_point_ct	11.76	0.0200	1.0K
KeccakF1600_StatePermute	5.88	0.0100	11.0K
keccak_inc_squeeze	0.00	0.0000	717.0K
shake256_inc_squeeze	0.00	0.0000	717.0K
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K

Function	%	self(s)	calls
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 82.4% of total

Function	%	self(s)	calls
mq_montymul	64.71	0.1100	8.4M
mq_add	11.76	0.0200	6.9M
mq NTT	5.88	0.0100	2.0K
mq_sub	0.00	0.0000	7.4M
mq_iNTT	0.00	0.0000	1.0K

Floating-Point / FFT — 0.0% of total

Function	%	self(s)	calls
mq_poly_sub	0.00	0.0000	1.0K

Other — 0.0% of total

Function	%	self(s)	calls
mq_rshift1	0.00	0.0000	9.0K
PQCLEAN_FALCON512_CLEAN_comp_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_crypto_sign_verify	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_is_short	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_modq_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_to_ntt_monty	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_verify_raw	0.00	0.0000	1.0K
do_verify	0.00	0.0000	1.0K
mq_poly_montymul_ntt	0.00	0.0000	1.0K
mq_poly_tomonty	0.00	0.0000	1.0K
run_falcon512_verify	0.00	0.0000	1

-03 build (total: 0.030 s)

Hash / PRNG — 33.3% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_hash_to_point_ct	33.33	0.0100	1.0K
keccak_inc_finalize	0.00	0.0000	718.0K
shake256_inc_squeeze	0.00	0.0000	717.0K
KeccakF1600_StatePermute	0.00	0.0000	11.0K
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	1.0K

Function	%	self(s)	calls
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 66.7% of total

Function	%	self(s)	calls
mq_montymul.constprop.0	33.33	0.0100	512.0K
mq_iNTT	33.33	0.0100	1.0K
mq_montymul	0.00	0.0000	7.9M
mq_sub	0.00	0.0000	7.4M
mq_add	0.00	0.0000	6.9M
mq_NTT	0.00	0.0000	2.0K

Floating-Point / FFT — 0.0% of total

Function	%	self(s)	calls
mq_poly_sub	0.00	0.0000	1.0K

Other — 0.0% of total

Function	%	self(s)	calls
mq_rshift1	0.00	0.0000	9.0K
PQCLEAN_FALCON512_CLEAN_comp_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_crypto_sign_verify	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_is_short	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_modq_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_to_ntt_monty	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_verify_raw	0.00	0.0000	1.0K
do_verify	0.00	0.0000	1.0K
mq_poly_montymul_ntt	0.00	0.0000	1.0K
mq_poly_tomonty	0.00	0.0000	1.0K
run_falcon512_verify	0.00	0.0000	1

4.4 Falcon-1024 — KeyGen

-00 build (total: 154.410 s)

Hash / PRNG — 3.0% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	1.96	3.0200	4.3M
keccak_inc_squeeze	0.87	1.3400	73.4M
shake256_inc_squeeze	0.11	0.1700	73.4M
get_rng_u64	0.09	0.1400	73.4M

Function	%	self(s)	calls
shake256_inc_finalize	0.00	0.0100	1.0K
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 33.5% of total

Function	%	self(s)	calls
modp_montymul	12.18	18.8100	2.84B
modp_add	5.57	8.6100	2.21B
zint_add_scaled_mul_small	3.31	5.1100	75.3M
zint_mod_small_unsigned	2.81	4.3400	113.6M
mkgauss	1.28	1.9700	36.7M
modp_NTT2_ext	1.17	1.8000	3.7M
zint_finish_mod	1.06	1.6400	1.9M
modp_R2	0.96	1.4900	7.5M
modp_sub	0.85	1.3200	899.5M
zint_add_mul_small	0.84	1.2900	49.7M
zint_co_reduce_mod	0.46	0.7100	965.1K
modp_iNTT2_ext	0.44	0.6800	3.5M
zint_norm_zero	0.38	0.5800	47.9M
zint_bezout	0.31	0.4800	1.1K
mq_montymul	0.29	0.4500	43.3M

Floating-Point / FFT — 62.5% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	32.35	49.9600	904.3M
FPR	10.96	16.9300	1.90B
PQCLEAN_FALCON1024_CLEAN_fpr_mul	10.62	16.4100	859.8M
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	3.18	4.9100	126.4M
fpr_ulsh	2.18	3.3600	904.3M
fpr_ursh	1.38	2.1300	904.3M
PQCLEAN_FALCON1024_CLEAN_FFT	0.46	0.7100	1.2M
PQCLEAN_FALCON1024_CLEAN_fpr_div	0.41	0.6400	5.8M
PQCLEAN_FALCON1024_CLEAN_iFFT	0.37	0.5700	608.1K
fpr_sub	0.19	0.3000	379.3M
fpr_rint	0.10	0.1500	9.2M
PQCLEAN_FALCON1024_CLEAN_poly_mul_fft	0.06	0.1000	1.2M
fpr_neg	0.06	0.0900	23.9M
fpr_of	0.05	0.0800	126.4M
fpr_sqr	0.02	0.0400	22.0M

Other — 1.0% of total

Function	%	self(s)	calls
poly_big_to_fp	0.16	0.2500	1.2M
modp_mkgm2	0.14	0.2200	1.9M
poly_sub_scaled	0.13	0.2000	1.1M
modp_div	0.13	0.2000	1.9M
fpr_lt	0.06	0.1000	14.2M
make_fg_step	0.06	0.0900	57.5K
solve_NTRU_intermediate	0.05	0.0800	8.3K
poly_sub_scaled_ntt	0.04	0.0700	56.2K
mq_div_12289	0.03	0.0500	1.2M
poly_small_sqnorm	0.03	0.0500	35.8K
PQCLEAN_FALCON1024_CLEAN_keygen	0.03	0.0500	1.0K
modp_R	0.03	0.0400	13.1M
modp_Rx	0.02	0.0300	1.8M
solve_NTRU_binary_depth0	0.02	0.0300	1.0K
modp_ninv31	0.02	0.0300	5.7M

-03 build (total: 34.260 s)

Hash / PRNG — 4.2% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	2.60	0.8900	4.1M
keccak_inc_finalize	0.90	0.3100	69.7M
get_rng_u64	0.58	0.2000	69.7M
shake256_inc_squeeze	0.09	0.0300	69.7M
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 40.7% of total

Function	%	self(s)	calls
modp_montymul	18.51	6.3400	2.80B
zint_add_scaled_mul_small	4.38	1.5000	75.2M
modp_add	3.30	1.1300	2.19B
zint_mod_small_unsigned	2.36	0.8100	112.2M
mkgauss	1.93	0.6600	34.8M
zint_rebuild_CRT.constprop.2	1.78	0.6100	76.4M
zint_finish_mod	1.34	0.4600	1.9M
zint_add_mul_small	1.31	0.4500	49.6M

Function	%	self(s)	calls
modp_sub	1.23	0.4200	894.5M
zint_co_reduce_mod	1.08	0.3700	955.6K
modp_iNTT2_ext	0.79	0.2700	3.5M
zint_bezout	0.67	0.2300	1.1K
zint_mod_small_signed	0.23	0.0800	62.6M
zint_co_reduce	0.23	0.0800	477.8K
zint_norm_zero	0.20	0.0700	45.7M

Floating-Point / FFT — 54.1% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	28.49	9.7600	865.4M
PQCLEAN_FALCON1024_CLEAN_fpr_mul	12.03	4.1200	828.2M
FPR	3.94	1.3500	1.82B
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	2.92	1.0000	124.7M
fpr_trunc	2.01	0.6900	865.4M
fpr_ursh	1.58	0.5400	865.4M
PQCLEAN_FALCON1024_CLEAN_fpr_div	1.23	0.4200	5.4M
PQCLEAN_FALCON1024_CLEAN_FFT	0.79	0.2700	1.2M
PQCLEAN_FALCON1024_CLEAN_iFFT	0.50	0.1700	600.8K
fpr_sub	0.23	0.0800	361.2M
fpr_rint	0.12	0.0400	9.1M
fpr_of	0.06	0.0200	124.7M
fpr_ulsh	0.06	0.0200	9.1M
fpr_neg	0.03	0.0100	22.6M
fpr_sqr	0.03	0.0100	19.6M

Other — 1.1% of total

Function	%	self(s)	calls
poly_big_to_fp	0.32	0.1100	1.2M
modp_div	0.15	0.0500	1.9M
solve_NTRU_intermediate	0.15	0.0500	8.3K
poly_sub_scaled	0.12	0.0400	1.1M
poly_sub_scaled_ntt	0.09	0.0300	56.3K
modp_mkkm2	0.06	0.0200	1.9M
make_fg_step.constprop.0	0.06	0.0200	47.7K
PQCLEAN_FALCON1024_CLEAN_poly_invnorm2_fft	0.03	0.0100	17.8K
make_fg	0.03	0.0100	9.4K
make_fg_step	0.03	0.0100	9.4K
PQCLEAN_FALCON1024_CLEAN_keygen	0.03	0.0100	1.0K
mq_montysqr	0.00	0.0000	12.9M
modp_R	0.00	0.0000	11.1M
modp_ninv31	0.00	0.0000	5.6M
fpr_inv	0.00	0.0000	5.4M

4.5 Falcon-1024 — Sign

-00 build (total: 36.090 s)

Hash / PRNG — 1.6% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_prng_refill	1.25	0.4500	76.6K
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct	0.19	0.0700	1.0K
prng_get_u8	0.06	0.0200	10.5M
prng_get_u64	0.06	0.0200	3.5M
keccak_inc_squeeze	0.00	0.0000	1.3M
shake256_inc_squeeze	0.00	0.0000	1.3M
KeccakF1600_StatePermute	0.00	0.0000	21.0K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
keccak_inc_finalize	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K
shake256_inc_ctx_release	0.00	0.0000	2.0K
shake256_inc_finalize	0.00	0.0000	2.0K

Integer Arithmetic — 1.8% of total

Function	%	self(s)	calls
mq_montymul	0.86	0.3100	44.0M
PQCLEAN_FALCON1024_CLEAN_gaussian0_sampler	0.50	0.1800	3.5M
mq_add	0.19	0.0700	20.5M
PQCLEAN_FALCON1024_CLEAN_sampler	0.11	0.0400	2.0M
mq NTT	0.08	0.0300	3.0K
mq_sub	0.03	0.0100	20.5M
mq_iNTT	0.00	0.0000	1.0K

Floating-Point / FFT — 96.3% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	49.99	18.0400	329.5M
FPR	17.54	6.3300	652.2M
PQCLEAN_FALCON1024_CLEAN_fpr_mul	15.36	5.5400	292.8M
fpr_ulsh	2.59	0.9400	329.5M
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	2.36	0.8500	23.8M
fpr_ursh	2.26	0.8100	336.5M
PQCLEAN_FALCON1024_CLEAN_fpr_div	1.62	0.5800	5.1M
PQCLEAN_FALCON1024_CLEAN_fpr_expm_p63	1.03	0.3700	3.5M
PQCLEAN_FALCON1024_CLEAN_FFT	0.53	0.1900	9.0K
PQCLEAN_FALCON1024_CLEAN_fpr_sqrt	0.44	0.1600	1.0M
fpr_sub	0.42	0.1500	157.4M

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_poly_LDL_fft	0.36	0.1300	1.0M
PQCLEAN_FALCON1024_CLEAN_poly_split_fft	0.28	0.1000	4.1M
PQCLEAN_FALCON1024_CLEAN_poly_mul_fft	0.25	0.0900	1.0M
fpr_trunc	0.19	0.0700	7.0M

Other — 0.4% of total

Function	%	self(s)	calls
mq_div_12289	0.08	0.0300	1.0M
PQCLEAN_FALCON1024_CLEAN_complete_private	0.08	0.0300	1.0K
BerExp	0.06	0.0200	3.5M
PQCLEAN_FALCON1024_CLEAN_comp_encode	0.06	0.0200	1.0K
fpr_floor	0.03	0.0100	2.0M
smallints_to_fpr	0.03	0.0100	8.0K
PQCLEAN_FALCON1024_CLEAN_trim_i8_decode	0.03	0.0100	3.0K
mq_montysqr	0.00	0.0000	11.3M
fpr_inv	0.00	0.0000	5.1M
mq_conv_small	0.00	0.0000	3.1M
fpr_irsh	0.00	0.0000	2.0M
mq_rshift1	0.00	0.0000	10.0K
PQCLEAN_FALCON1024_CLEAN_crypto_sign_signature	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_is_short_half	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_sign_dyn	0.00	0.0000	1.0K

-03 build (total: 7.570 s)

Hash / PRNG — 1.1% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_prng_refill	0.66	0.0500	76.6K
prng_get_u8	0.13	0.0100	10.5M
prng_get_u64	0.13	0.0100	3.5M
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct	0.13	0.0100	1.0K
keccak_inc_finalize	0.00	0.0000	1.3M
shake256_inc_squeeze	0.00	0.0000	1.3M
KeccakF1600_StatePermute	0.00	0.0000	21.0K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K
shake256_inc_ctx_release	0.00	0.0000	2.0K
shake256_inc_finalize	0.00	0.0000	2.0K
shake256_inc_init	0.00	0.0000	2.0K

Integer Arithmetic — 3.2% of total

Function	%	self(s)	calls
mq_montymul	1.06	0.0800	42.0M
PQCLEAN_FALCON1024_CLEAN_gaussian0_sampler	0.66	0.0500	3.5M
mq_add	0.26	0.0200	20.5M
mq_sub	0.26	0.0200	20.5M
PQCLEAN_FALCON1024_CLEAN_sampler	0.26	0.0200	2.0M
mq_montymul.constprop.0	0.26	0.0200	2.0M
mq NTT	0.26	0.0200	3.0K
mq_iNTT	0.13	0.0100	1.0K

Floating-Point / FFT — 95.5% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	53.24	4.0300	329.5M
PQCLEAN_FALCON1024_CLEAN_fpr_mul	17.04	1.2900	292.8M
PQCLEAN_FALCON1024_CLEAN_fpr_div	5.55	0.4200	5.1M
FPR	4.62	0.3500	652.2M
fpr_trunc	2.77	0.2100	336.5M
fpr_ursh	2.64	0.2000	336.5M
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	2.38	0.1800	23.8M
PQCLEAN_FALCON1024_CLEAN_fpr_expm_p63	1.72	0.1300	3.5M
PQCLEAN_FALCON1024_CLEAN_FFT	1.19	0.0900	9.0K
fpr_half	0.66	0.0500	36.9M
PQCLEAN_FALCON1024_CLEAN_poly_split_fft	0.66	0.0500	4.1M
PQCLEAN_FALCON1024_CLEAN_fpr_sqrt	0.66	0.0500	1.0M
fpr_sub	0.40	0.0300	157.4M
PQCLEAN_FALCON1024_CLEAN_poly_merge_fft	0.40	0.0300	2.0M
ffSampling_fft_dyntree.constprop.0	0.40	0.0300	1.0K

Other — 0.3% of total

Function	%	self(s)	calls
BerExp	0.13	0.0100	3.5M
PQCLEAN_FALCON1024_CLEAN_comp_encode	0.13	0.0100	1.0K
mq_montysqr	0.00	0.0000	11.3M
fpr_inv	0.00	0.0000	5.1M
mq_conv_small	0.00	0.0000	3.1M
fpr_floor	0.00	0.0000	2.0M
fpr_irsh	0.00	0.0000	2.0M
mq_div_12289	0.00	0.0000	1.0M
mq_rshift1	0.00	0.0000	10.0K
smallints_to_fpr	0.00	0.0000	8.0K
PQCLEAN_FALCON1024_CLEAN_trim_i8_decode	0.00	0.0000	3.0K
PQCLEAN_FALCON1024_CLEAN_complete_private0	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_crypto_sign_signature0	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_is_short_half	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_sign_dyn	0.00	0.0000	1.0K

4.6 Falcon-1024 — Verify

-00 build (total: 0.350 s)

Hash / PRNG — 17.1% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct1	14.29	0.0500	1.0K
KeccakF1600_StatePermute	2.86	0.0100	20.0K
keccak_inc_squeeze	0.00	0.0000	1.3M
shake256_inc_squeeze	0.00	0.0000	1.3M
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 80.0% of total

Function	%	self(s)	calls
mq_montymul	45.71	0.1600	18.4M
mq_add	14.29	0.0500	15.4M
mq_NTT	8.57	0.0300	2.0K
mq_sub	5.71	0.0200	16.4M
mq_iNTT	5.71	0.0200	1.0K

Floating-Point / FFT — 2.9% of total

Function	%	self(s)	calls
mq_poly_sub	2.86	0.0100	1.0K

Other — 0.0% of total

Function	%	self(s)	calls
mq_rshift1	0.00	0.0000	10.0K
PQCLEAN_FALCON1024_CLEAN_comp_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_crypto_sign_verify	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_is_short	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_modq_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_to_ntt_monty	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_verify_raw	0.00	0.0000	1.0K
do_verify	0.00	0.0000	1.0K
mq_poly_montymul_ntt	0.00	0.0000	1.0K
mq_poly_tomonty	0.00	0.0000	1.0K
run_falcon1024_verify	0.00	0.0000	1

-03 build (total: 0.150 s)

Hash / PRNG — 20.0% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct	20.00	0.0300	1.0K
keccak_inc_finalize	0.00	0.0000	1.3M
shake256_inc_squeeze	0.00	0.0000	1.3M
KeccakF1600_StatePermute	0.00	0.0000	20.0K
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 73.3% of total

Function	%	self(s)	calls
mq_montymul	53.33	0.0800	17.4M
mq_sub	6.67	0.0100	16.4M
mq_add	6.67	0.0100	15.4M
mq_montymul.constprop.0	6.67	0.0100	1.0M
mq_NTT	0.00	0.0000	2.0K
mq_iNTT	0.00	0.0000	1.0K

Floating-Point / FFT — 0.0% of total

Function	%	self(s)	calls
mq_poly_sub	0.00	0.0000	1.0K

Other — 6.7% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_is_short	6.67	0.0100	1.0K
mq_rshift1	0.00	0.0000	10.0K
PQCLEAN_FALCON1024_CLEAN_comp_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_crypto_sign_verify	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_modq_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_to_ntt_monty	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_verify_raw	0.00	0.0000	1.0K
do_verify	0.00	0.0000	1.0K
mq_poly_montymul_ntt	0.00	0.0000	1.0K
mq_poly_tomonty	0.00	0.0000	1.0K
run_falcon1024_verify	0.00	0.0000	1

5 Key Findings

- **KeyGen is dominated by Integer NTT and FP/FFT arithmetic.** `modp_montymul` (Montgomery multiplication for NTT) and `fpr_add` (FFT floating-point add) together account for over 40% of total keygen time. The lattice basis generation (`zint_*`, `mkgauss`) contributes a further $\sim 10\text{--}15\%$.
- **Sign is almost entirely FP/FFT.** The `ffSampling_fft_dyntree` recursive sampler drives repeated calls to `fpr_add` ($\sim 50\%$) and `fpr_mul` ($\sim 15\%$). Integer NTT (`mq_montymul`) is visible but minor ($< 2\%$).
- **Verify is entirely Hash and Integer NTT.** No floating-point FFT is used. Time splits between `hash_to_point_ct` (SHAKE-256 hashing of the message) and `mq_montymul` / `mq NTT` (NTT-based polynomial multiplication over $\mathbb{Z}_q$). This is why verify is $70\text{--}300\times$ faster than keygen.
- **Falcon-1024 costs $\approx 2\text{--}3\times$ more than Falcon-512.** The FFT and NTT algorithms are $\mathcal{O}(N \log N)$; doubling N from 512 to 1024 roughly doubles the cost with an additional $\log$ factor.
- **O3 gives $4\text{--}5\times$ speedup on keygen/sign, $7\times$ on verify.** The FPR constructor overhead drops from $\sim 15\%$ to $< 5\%$ under O3 (compiler promotes the struct boxing). Verify benefits most from O3 because `mq_montymul` vectorises well.